

A 500 Gbps Ultra-Compact Silicon Photonic Mach-Zehnder Modulator with >110 GHz Bandwidth

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Abstract *We demonstrate a 500 Gbps silicon modulator with a length of 300 μm , integrating chirped phase-shifted grating and electro-optic frequency-domain equalization. The device achieves a tuning efficiency of 66 pm/V and an electro-optic bandwidth exceeding 110 GHz.*

Introduction

With the rapid growth of emerging applications such as artificial intelligence and cloud computing, interconnect traffic in data centre and high-performance computing (HPC) systems has increased dramatically, imposing ever more stringent requirements on bandwidth density and energy efficiency. Conventional copper interconnects, constrained by severe frequency-dependent loss and high-power consumption, are increasingly unable to support next-generation high-capacity interconnect architectures. Benefiting from CMOS compatibility, high integration density, and low power consumption, silicon photonics has become a promising platform for realizing high-performance on-chip and board-level optical interconnects [1]. As a key component in optical interconnect links, the bandwidth, energy efficiency, and footprint of silicon photonic modulators directly determine the transmission capability and integration density of the overall system [2].

However, to satisfy the demands of single-lane transmission at 400 Gbps and beyond, as well as future 3.2T/6.4T optical modules, the bandwidth of conventional silicon photonic modulators is approaching its fundamental limit. For modulators based on the plasma-dispersion effect, performance is typically constrained by carrier response dynamics, parasitic RC effects, and the velocity mismatch between microwave and optical waves, making further extension of the electro-optic bandwidth highly challenging. Although reducing the device length can partially mitigate RC limitations, it inevitably degrades modulation efficiency while increasing the required driving voltage and power consumption. Therefore, overcoming the traditional trade-off among bandwidth, efficiency, and power consumption, and developing new bandwidth-enhancement strategies, is of critical importance for next-generation high-speed silicon photonic modulators. Among various modulators, Mach-Zehnder modulator (MZM), owing to its high linearity and broadband operation capability, has become one of the most widely adopted solutions

for high-speed optical interconnects. Numerous studies have focused on it through various approaches, including impedance peaking [3], slow light enhancement [4,5], electrode segmentation [6], heterogeneous integration platform [7] and so on.

In this work, we present a high-speed silicon modulator at O-band that employs a slow-light-enhanced chirped phase-shift grating (CPSG) and electro-optic frequency-domain equalization (EOFDE) to enhance both the modulation bandwidth and efficiency. The device exhibits an EO bandwidth exceeding 110 GHz and a tuning efficiency of 66 pm/V. Furthermore, 500 Gbps PAM4 transmission is experimentally demonstrated. This approach provides a promising solution for next-generation ultrahigh-speed optical interconnects in data centre and high-performance computing applications.

Design and fabrication

Fig. 1(a) illustrates the structure of the proposed modulator. By integrating the CPSG with EOFDE, an architecture that simultaneously enhances tuning efficiency and EO bandwidth is realized. This approach effectively mitigates the high-frequency roll-off of the silicon photonic modulator without compromising tuning efficiency, while also enabling a large optical bandwidth. The CPSG is employed to regulate the optical field propagation and phase response in the phase-shifter arms. By introducing a chirped grating distribution along the propagation direction and incorporating multiple phase-shift sections, the device exhibits a controllable transmission spectrum and group-delay response around the target wavelength. Compared with a uniform grating, the CPSG provides greater design flexibility in engineering the spectral position, bandwidth, phase response, while also enabling a larger product of group index and optical bandwidth. The EOFDE comprises a main modulation section, a waveguide crossing, and a modulation section with opposite polarities, enabling weighted equalization of different frequency com-

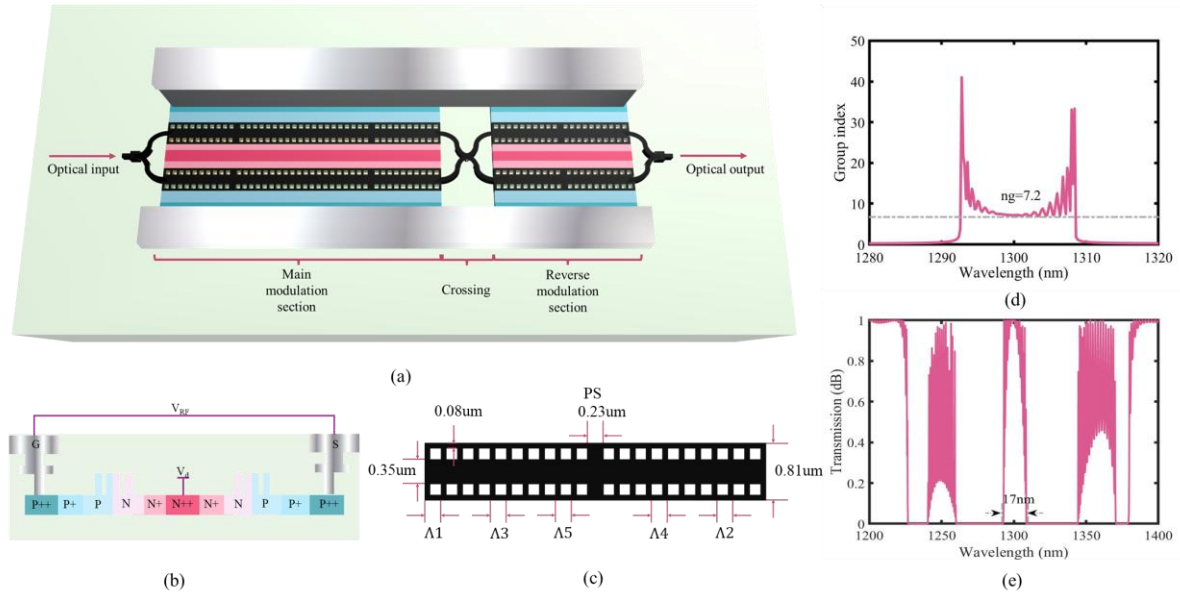


Fig. 1: (a) Schematic of the silicon modulator integrating CPSG and EOFDE. (b) Cross-sectional view of the modulator. (c) Top view of the CPSG. (d) Group index of the CPSG. (e) Transmission spectrum of the CPSG

ponents by adjusting the relative proportions of these sections. Through the synergistic operation of CPSG and EOFDE in both optical and electrical domains, the overall frequency response is extended, resulting in improved in-band flatness.

Fig. 1(b) shows the cross-sectional structure of the modulator. The device is fabricated on a silicon-on-insulator (SOI) platform, with a 3-μm buried oxide layer and a 220-nm top silicon layer. The etch depth is 130 nm. As shown in Fig. 1(c), the main waveguide adopts a rib-based CPSG structure containing 20 phase-shift (PS) sections. Each PS section has a waveguide width of 0.81 μm and a length of 0.23 μm. Grating units with progressively varying periods ($\Lambda_1 = 0.23 \mu\text{m}$, $\Lambda_2 = 0.232 \mu\text{m}$, $\Lambda_3 = 0.233 \mu\text{m}$, $\Lambda_4 = 0.235 \mu\text{m}$, $\Lambda_5 = 0.236 \mu\text{m}$) and duty cycles (0.348, 0.349, 0.352, 0.353, 0.355) are symmetrically arranged on both sides. Each grating unit is repeated twice and realized as two etched slots in the rib regions of the waveguide. This design enhances the effective refractive index of the waveguide without compromising the optical bandwidth. Fig. 1(d) and (e) show the group refractive index and transmission of the CPSG, exhibiting a group index of 7.2 and an optical bandwidth of 17 nm. The modulation region operates based on the plasma-dispersion effect in a PN junction. The doping profile is optimized using a multi-level design to balance modulation efficiency, junction capacitance, series resistance, and excess optical loss. Specifically, heavily doped regions ($5 \times 10^{20} \text{ cm}^{-3}$) are employed to form low-resistance ohmic contacts, while the moderately doped p^+ and n^+ regions are set to $3 \times 10^{18} \text{ cm}^{-3}$ to achieve an optimal trade-off among optical loss, junction-limited bandwidth, and microwave attenuation. A traveling-wave (TW) ground-signal (GS) electrode configuration

with a 50-Ω termination resistor is employed and co-optimized with the EOFDE to enhance the frequency response. In addition, thermo-optic phase shifters are integrated into both interferometer arms to enable precise bias control, ensuring operation at the optimum linear bias point.

Experiment results

Fig. 2(a) shows the transmission spectra at 0 V and 6 V, indicating an insertion loss of 6 dB and a tuning efficiency of approximately 66 pm/V, corresponding to a $V_{\pi} \cdot L$ of 0.9 V·cm. The small-signal S21 measurement was performed using a 110 GHz bandwidth optical vector network analyser combined with a 100 GHz photodetector. Optical input/output of the device was realized through grating couplers, while the RF signal was applied to the device via a 67 GHz GS probe. A bias tee was used to apply a reverse bias voltage to the modulator. As shown in Fig. 2(b), the EO response of the modulator remains flat over 0-110 GHz frequency range under a -1 V bias, with negligible roll-off, indicating excellent broadband electro-optic performance.

Fig. 2(c) illustrates the schematic of the large-signal experimental setup. The RF signal was generated by a 256 GSa/s arbitrary waveform generator (Keysight M8199B), amplified by a 67 GHz, 11 dB amplifier (SHF M827B), and then fed into the modulator through a 67 GHz GS RF probe. The modulated optical output was amplified by a praseodymium-doped fiber amplifier (PDFA) and then detected by a broadband oscilloscope (Keysight N1000A). Fig. 2(d-i) present the optical eye diagrams of the proposed modulator for PRBS7 at different data rates. For NRZ signals at 128 Gbps and 192 Gbps, the measured

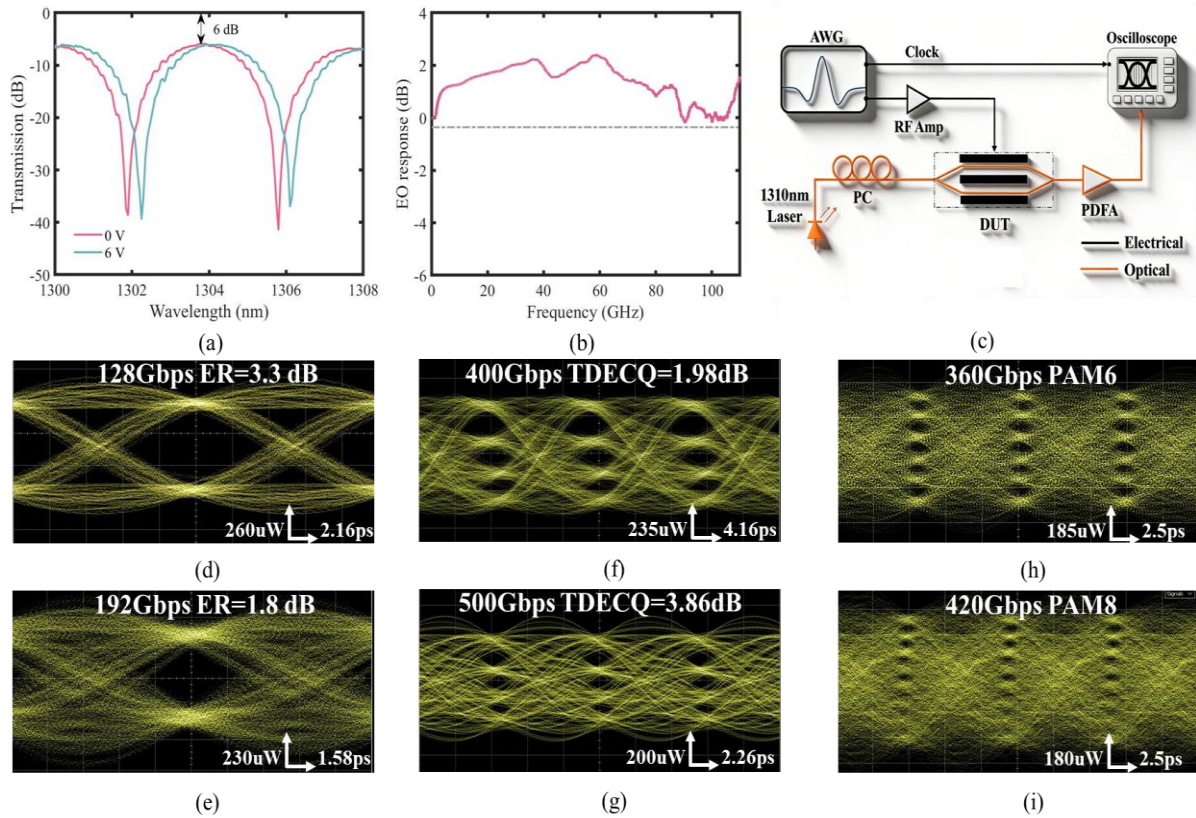


Fig. 2: (a) Measured transmission spectra of the modulator at 0 V and 6 V bias. (b) EO response of the modulator at -1 V bias. (c) Experimental setup for eye diagram measurements of the modulator. Measured optical eye diagrams at data rates of (d) 128 Gbps NRZ, (e) 192 Gbps NRZ, (f) 400 Gbps PAM4, (g) 500 Gbps PAM4, (h) 360 Gbps PAM6, and (i) 420 Gbps PAM8.

extinction ratios (ER) were 3.3 dB and 1.8 dB, respectively. For PAM4 signals at 400 Gbps and 500 Gbps, the output eye diagrams also exhibited clear multilevel separation, with transmitter dispersion eye closure quaternary (TDECQ) values of 1.98 dB and 3.86 dB, respectively. The eye diagrams for PAM6 at 360 Gbps and PAM8 at 420 Gbps also show well-defined multilevel separations, further verifying the capability of the proposed modulator to support higher-order modulation formats.

The experimental results show that at 400 Gbps, the device exhibits a clearly open eye diagram with stable crossing points and a relatively high extinction ratio, demonstrating strong high-speed modulation capability. When the data rate is increased to 500 Gbps, the eye opening becomes partially compressed compared with lower-speed operation due to factors such as driver-chain loss, limited test system bandwidth, and accumulated high-frequency noise. Nevertheless, a discernible open eye is maintained, indicating good potential for ultra-high-speed transmission. Overall, these results confirm that the proposed modulator offers significant advantages in both bandwidth performance and high-speed modulation capability.

Conclusions

This paper proposes a tuning-efficiency and EO-bandwidth enhancement scheme for a high-speed silicon modulator by integrating CPSG and EOFDE. By leveraging CPSG to precisely tailor optical field propagation and phase response, and employing EOFDE for weighted compensation of different frequency components, the proposed approach effectively mitigates the high-frequency roll-off of silicon photonic modulators, significantly improving both the electro-optic bandwidth and in-band response flatness. Experimental results show that the modulator achieves a 110 GHz zero-roll-off bandwidth, a tuning efficiency of 66 pm/V, and supports 500 Gbps PAM4 transmission, demonstrating excellent performance for ultra-high-speed optical interconnect applications. This scheme provides a promising technical pathway toward ultra-high-speed, large-bandwidth silicon photonic interconnects for next-generation data centers and high-performance computing systems.

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